

# Two-sample reading budget: same expert and context packing

CFPerm prototype

Not TMLE and not causal.  $W$  is a batch tag;  $\hat{e} \approx P(W=1 \mid X)$ .  $\pi$  is a shrinkage prior over which human block localizes the two-sample difference. It is not a treatment propensity and is not used to target  $Y$ .

## 1 Same expert versus packing

A *same expert* is  $\hat{m} = \arg \max \pi$ , then  $f \in \mathcal{G}_{\hat{m}}$  (functions of  $X_{B_{\hat{m}}}$  only) for every query. That template does not depend on  $x$ . An instance gate  $\hat{m}(x) = \arg \max s_m(x)$  is a different object.

Context packing is one reader on concatenated raw blocks  $E = E(\pi)$ , not an opinion pool of  $\hat{e}_m$ . Disabled channels are omitted, not emitted as empty headers. Abstain means the gap is not localized: do not drop a block (pack all, or HITL). Forced top-1 on a diffuse shift discards signal.

## 2 Specialist $\hat{e}_m$ (one block, all rows)

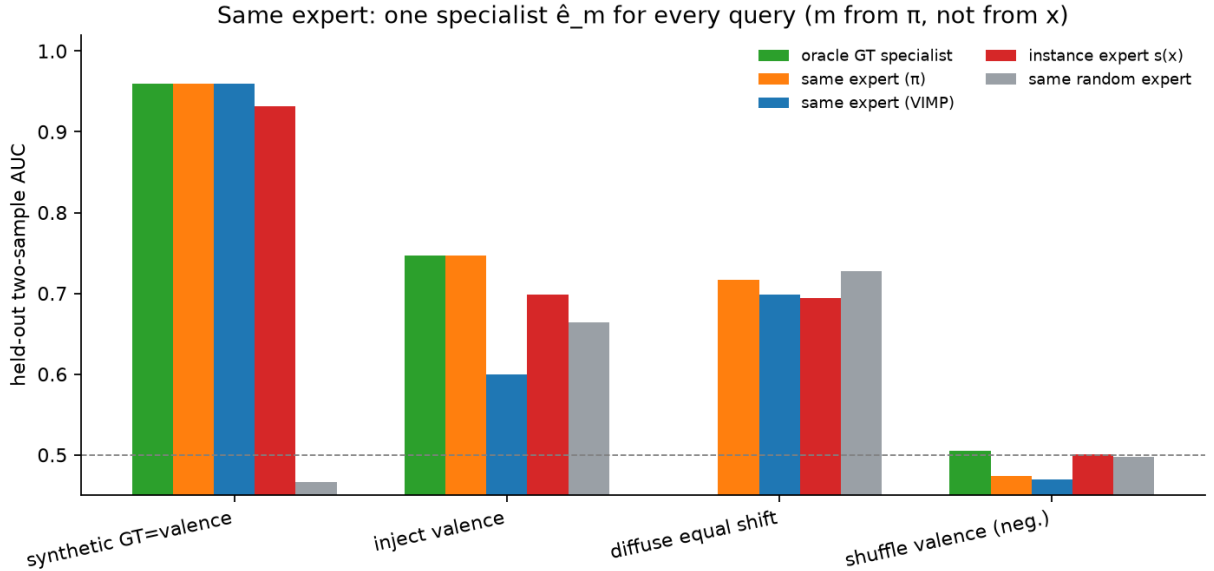


Figure 1: Frozen specialist AUC.  $\pi$  picks one  $m$  for the whole test set.

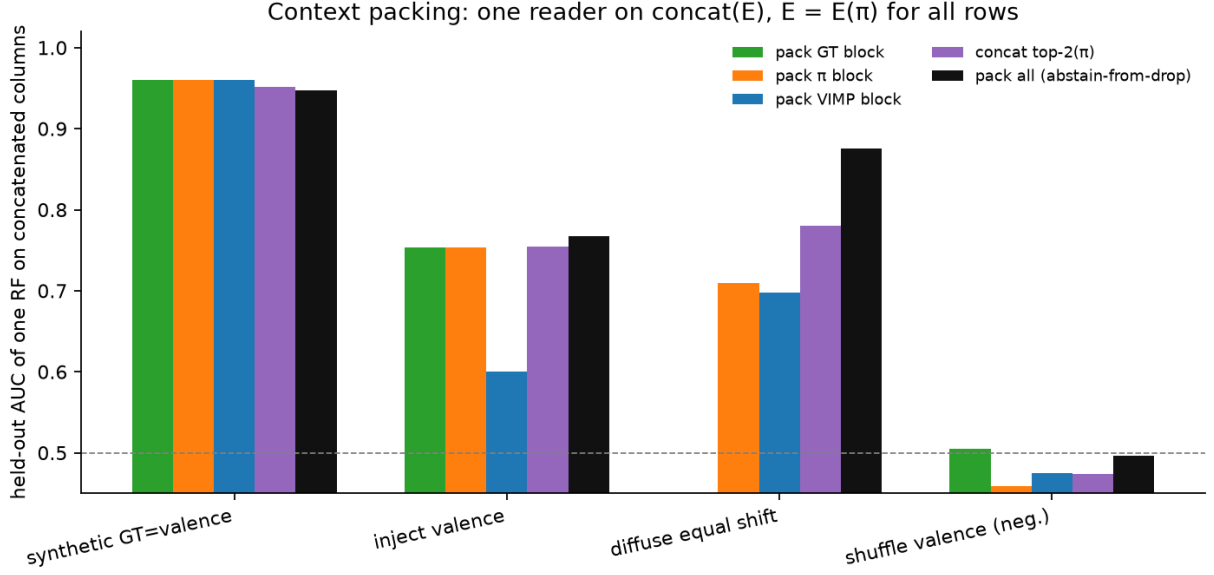


Figure 2: Column-concat of  $E(\pi)$ . Pack-all is the abstain-from-dropping action.

### 3 Packed reader (one RF on concat $E$ )

Inject valence with a wide text block: VIMP hires the wide specialist;  $\pi$  hires valence and the packed reader matches the GT block. Diffuse equal shift: pack-all beats one specialist. Shuffle valence: hit collapses.

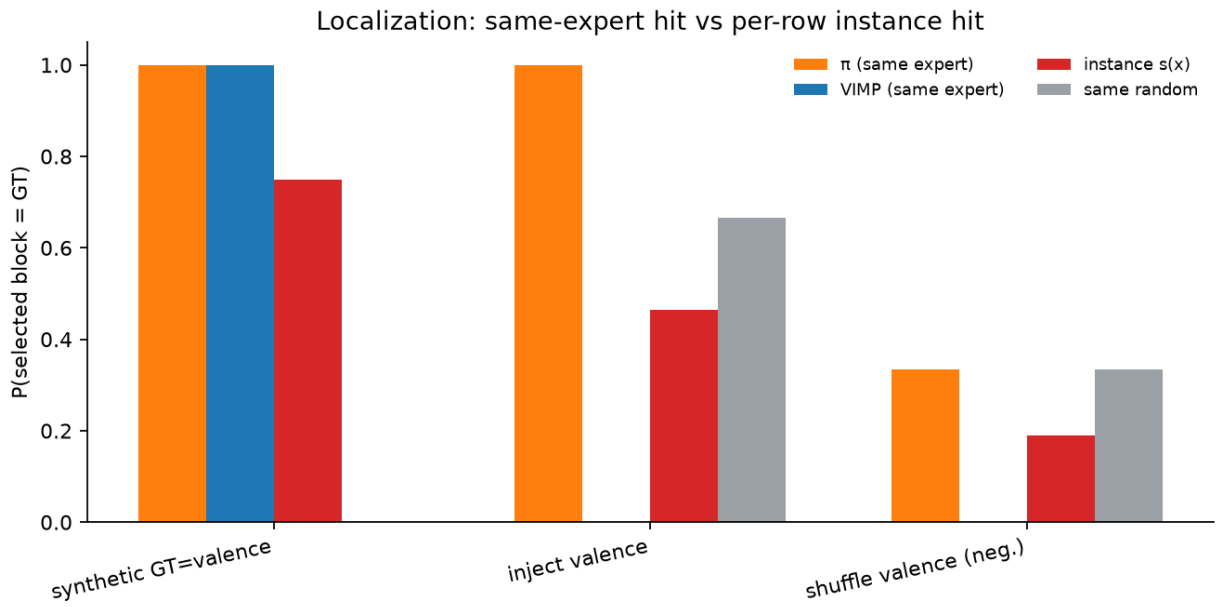


Figure 3: Localization hit. Same-expert hit is  $1\{\hat{m} = m^*\}$ , not a per-row mix.